Data Quality Audit and Preliminary Analysis

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Loading packages

```
library(haven)
library(tidyverse)
```

Import dataset

```
All_Country <- read_csv("Data/Processed/AllCountries_Processed.csv")
```

Dataset Structure & Summary

```
dim(All_Country)
```

```
## [1] 19449    25
```

```
summary(All_Country)
```

```
##    Country             HH_ID           Subdivision1       Subdivision2
##  Length:19449       Length:19449       Length:19449       Length:19449
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##
##   UrbanOrRural   ElectrificationStatus HouseholdSize     DwellingType
##  Min.   :1.000   Length:19449          Min.   : 1.000   Min.   :  1.000
##  1st Qu.:1.000   Class :character      1st Qu.: 3.000   1st Qu.:  1.000
##  Median :2.000   Mode  :character      Median : 4.000   Median :  1.000
##  Mean   :1.508                         Mean   : 4.777   Mean   :  2.675
##  3rd Qu.:2.000                         3rd Qu.: 6.000   3rd Qu.:  2.000
##  Max.   :3.000                         Max.   :23.000   Max.   :555.000
##  NA's   :3297                          NA's   :7119     NA's   :292
##  DwellingOwnership DwellingRoomsNumber  BankAccount    GridConnected
##  Min.   :1.000     Min.   :   0.000    Min.   :  1.0   Min.   :  0.000
##  1st Qu.:1.000     1st Qu.:   2.000    1st Qu.:  1.0   1st Qu.:  1.000
##  Median :1.000     Median :   3.000    Median :  2.0   Median :  2.000
##  Mean   :1.278     Mean   :   3.364    Mean   : 23.4   Mean   :  1.859
##  3rd Qu.:2.000     3rd Qu.:   4.000    3rd Qu.:  2.0   3rd Qu.:  2.000
##  Max.   :2.000     Max.   :2000.000    Max.   :111.0   Max.   :900.000
##  NA's   :289       NA's   :3925        NA's   :287     NA's   :3810
##  NumYearsGridConnected GridConnectionFeeLocal GridConnectionFeeUSD
```

```
##  Min.   :     0.0      Min.   :    -8         Min.   :  -0.684
##  1st Qu.:     2.0      1st Qu.:     0         1st Qu.:   0.000
##  Median :     4.0      Median :  1000         Median :  14.706
##  Mean   :   312.1      Mean   : 25249         Mean   :  76.256
##  3rd Qu.:    10.0      3rd Qu.: 35000         3rd Qu.:  70.457
##  Max.   :100000.0      Max.   :800000         Max.   :6862.745
##  NA's   :12616         NA's   :14453          NA's   :14453
##  InternalWiringFeeLocal InternalWiringFeeUSD ElectricityBillPriceLocal
##  Min.   :    -8         Min.   :  -0.684     Min.   :   -8
##  1st Qu.:  4500         1st Qu.:  18.541     1st Qu.:  500
##  Median : 12000         Median :  37.083     Median : 1500
##  Mean   : 21943         Mean   :  68.980     Mean   : 3626
##  3rd Qu.: 25250         3rd Qu.:  75.472     3rd Qu.: 4000
##  Max.   :700000         Max.   :1698.514     Max.   :99000
##  NA's   :16765          NA's   :16765        NA's   :14129
##  ElectricityBillPriceUSD ElectricityConsumptionkWh OutageDurationHours
##  Min.   : -0.684          Min.   :   -8.00          Min.   :  -8.0
##  1st Qu.:  2.830          1st Qu.:    1.00          1st Qu.:   2.0
##  Median :  4.944          Median :    4.65          Median :   5.0
##  Mean   :  9.531          Mean   :  138.22          Mean   : 102.6
##  3rd Qu.: 10.029          3rd Qu.:   39.40          3rd Qu.:  54.5
##  Max.   :392.157          Max.   :67115.00          Max.   :3300.0
##  NA's   :14129            NA's   :11606             NA's   :14342
##  MobilePhonesOwned  WillingToPayForConnection
##  Min.   :       0   Min.   :1.000
##  1st Qu.:       1   1st Qu.:1.000
##  Median :       1   Median :2.000
##  Mean   :   24392   Mean   :1.533
##  3rd Qu.:       2   3rd Qu.:2.000
##  Max.   :97985235   Max.   :4.000
##  NA's   :7875       NA's   :10680
##  EstimateOfElectricitySpendingLocal EstimateOfElectricitySpendingUSD
##  Min.   :    0                      Min.   :  0.000
##  1st Qu.:    2                      1st Qu.:  0.021
##  Median :  250                      Median :  1.961
##  Mean   : 1316                      Mean   :  4.920
##  3rd Qu.: 1000                      3rd Qu.:  4.902
##  Max.   :50000                      Max.   :490.196
##  NA's   :14734                      NA's   :14734
```

Dimension

Our dataset has 19449 rows and 25 columns. It is in long format, and each row represents a household.

Error

From the summary function, we can see that the grid connection fee, internal wiring fee, electricity bill price, electricity consumption, and outage duration all have a minimum value of -8. By investigating the dataset, all of these "-8" values come from Zambia, so we might need to go back to the original dataset to debug.

The "Grid connected" column for Liberia is a value ranging from 3-900 while the other countries have values of 1 or 2 where 1 means "yes" and 2 means "no". Need to verify if we used the correct columns from the Liberia dataset.

The "Outage duration hours" column for Liberia is also very large, with a max of 3300 hours. This number should be the outage duration hours in a week, and a week only has 168 hours. Need to go back to the

Liberia dataset to check if the column is correct, and if the unit is in hours or minute.

"Electricity consumption kwh" column for Niger has a max of 67115.00. This column should be electricity consumption kwh per month. Niger in general has large numbers which is unreasonable for a household to use this much electricity. Therefore, might need to double check the unit or see if these are extreme values.

Formatting

The electrification status are supposed in numeric values representing different categories, however this column for Niger is "character" with a written description.

Extreme value

Dwelling room number has a max value of 2000 in Liberia which is unreasonable. Mobile phone number has a max value of 97985235 in Niger which is also unreasonable and need to be removed.

Missing Value

```
colSums(is.na(All_Country))
```

```
##                            Country                              HH_ID
##                                  0                                 17
##                        Subdivision1                        Subdivision2
##                                  2                                  2
##                        UrbanOrRural                ElectrificationStatus
##                                3297                              10811
##                       HouseholdSize                       DwellingType
##                                7119                                292
##                    DwellingOwnership                  DwellingRoomsNumber
##                                 289                               3925
##                         BankAccount                       GridConnected
##                                 287                               3810
##                NumYearsGridConnected               GridConnectionFeeLocal
##                               12616                              14453
##                 GridConnectionFeeUSD               InternalWiringFeeLocal
##                               14453                              16765
##                 InternalWiringFeeUSD            ElectricityBillPriceLocal
##                               16765                              14129
##              ElectricityBillPriceUSD            ElectricityConsumptionkWh
##                               14129                              11606
##                  OutageDurationHours                   MobilePhonesOwned
##                               14342                               7875
##           WillingToPayForConnection EstimateOfElectricitySpendingLocal
##                               10680                              14734
##    EstimateOfElectricitySpendingUSD
##                               14734
```

```
HH_ID_NA <- All_Country %>% filter(is.na(HH_ID))
#view(HH_ID_NA)
```

Our Dataset has lots of NAs especially for the columns asking about more details of electricity bills, connection fees and outages, etc.

There are 17 NAs in the HH_ID column. After viewing the rows with a missing household ID, 16 rows are from Liberia and 1 is from Niger. The other columns of these rows that has a missing household ID are also

mostly NAs. We might consider removing these rows from the dataset because we can't be sure these are data from unique households.

Duplicate

```
Duplicate <- All_Country %>%
  filter(!is.na(HH_ID)) %>%
  add_count(Country, HH_ID, name = "count") %>%
  filter(count > 1) %>%
  arrange(Country, HH_ID)
#view(Duplicate)
```

We looked for duplicates in the HH_ID column, because each household ID **should** be different. However, lots of duplicates came up, and the household size and dwelling type are different, even with the same household ID. Therefore the different rows with the same household ID should still be different households and we will keep these data.

Preliminary Analysis

```
library(dplyr)
library(ggplot2)
```

```
# Calculating sample size by country
country_summary <- All_Country %>%
  group_by(Country) %>%
  summarise(
    Households = n(),
    .groups = "drop"
  ) %>%
  mutate(
    Share = round(Households/sum(Households)*100,1)
  )

country_summary
```

```
## # A tibble: 5 x 3
##   Country Households Share
##   <chr>        <int> <dbl>
## 1 Kenya         4590  23.6
## 2 Liberia       3903  20.1
## 3 Niger         4049  20.8
## 4 Rwanda        3295  16.9
## 5 Zambia        3612  18.6
```

As noted in the data quality audit, the final dataset contains 19,449 household observations across five countries: Kenya, Liberia, Niger, Rwanda, and Zambia. The table above shows the sample size of each country in the final dataset and what share of observations each country accounts for. Kenya has the largest share of observations in the dataset, with a sample size of 4590 households represented (23.6%) while Zambia has the lowest number of observations with a smaple size of 3612 households (18.6%). These differences are small enough that the dataset is relatively balanced across countries. This provides a reasonably even basis for comparing household-level patterns across the five countries, without any single country overwhelmingly dominating the pooled dataset.

Analysis 1: Grid Connection

```
# Calculating grid connection rates
grid_connection_sumary <- All_Country %>%
  filter(GridConnected %in% c(1,2)) %>%
  group_by(Country) %>%
  summarise(
    Total_Valid = n(),
    Connected = sum(GridConnected == 1),
    Not_Connected = sum(GridConnected == 2),
    Connection_Rate = round(
      mean(GridConnected == 1)*100,1
    ),
    .groups = "drop"
  )

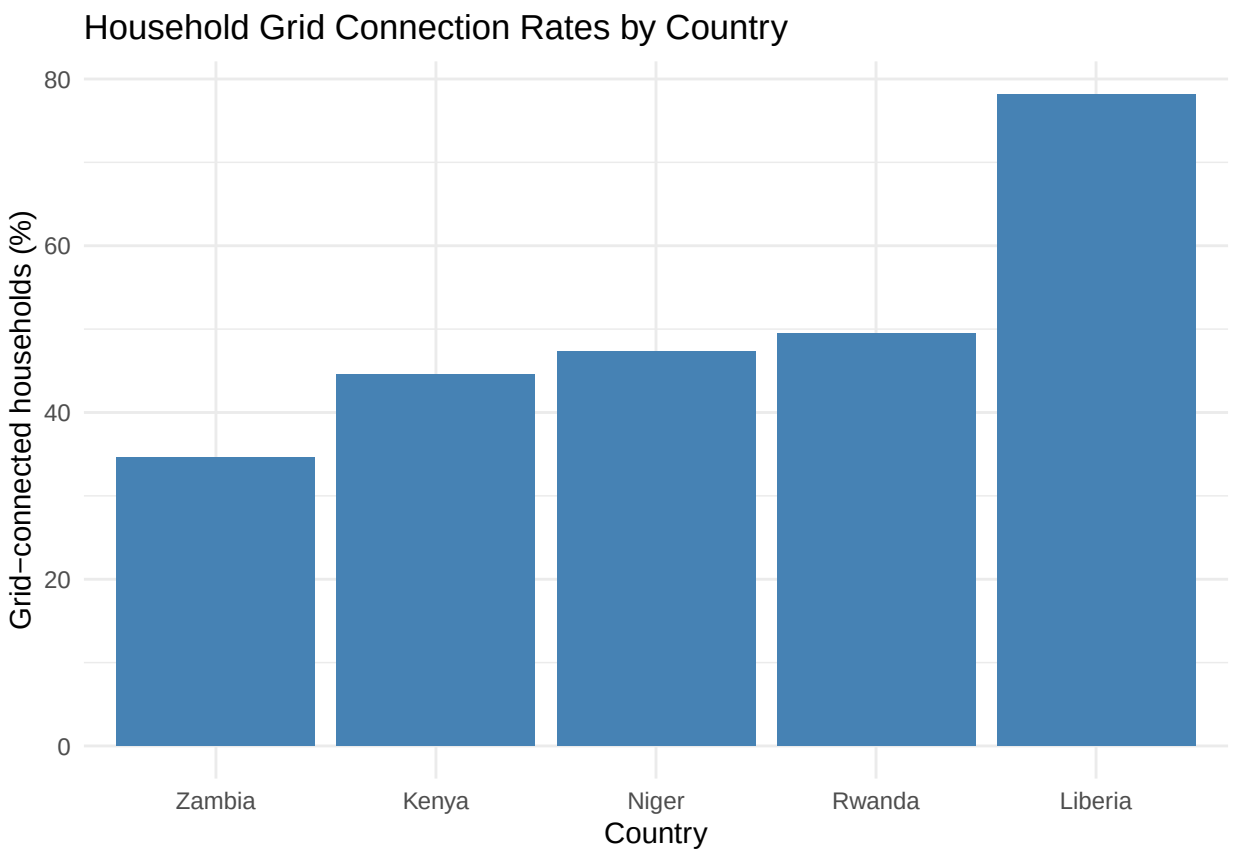
grid_connection_sumary
```

```
## # A tibble: 5 x 5
##   Country Total_Valid Connected Not_Connected Connection_Rate
##   <chr>         <int>     <int>         <int>           <dbl>
## 1 Kenya          4590      2043          2547            44.5
## 2 Liberia          87        68            19            78.2
## 3 Niger          4022      1901          2121            47.3
## 4 Rwanda         3295      1632          1663            49.5
## 5 Zambia         3537      1224          2313            34.6
```

The table shows substantial variation in electricity connection rates across the five represented countries. Kenya has 4590 valid observations with over 44.5% households with grid connection. Niger and Rwanda have similar connection rates of 47.3% and 49.5%, respectively, while Zambia has the lowest connection rate of 34.6%. Liberia has the highest connection rate of 78.2%, but this is based on only 87 valid observations, as compared to several thousand in the other countries. Therefore, Liberia's results should be interpreted cautiously.

The figure below visualizes this result:

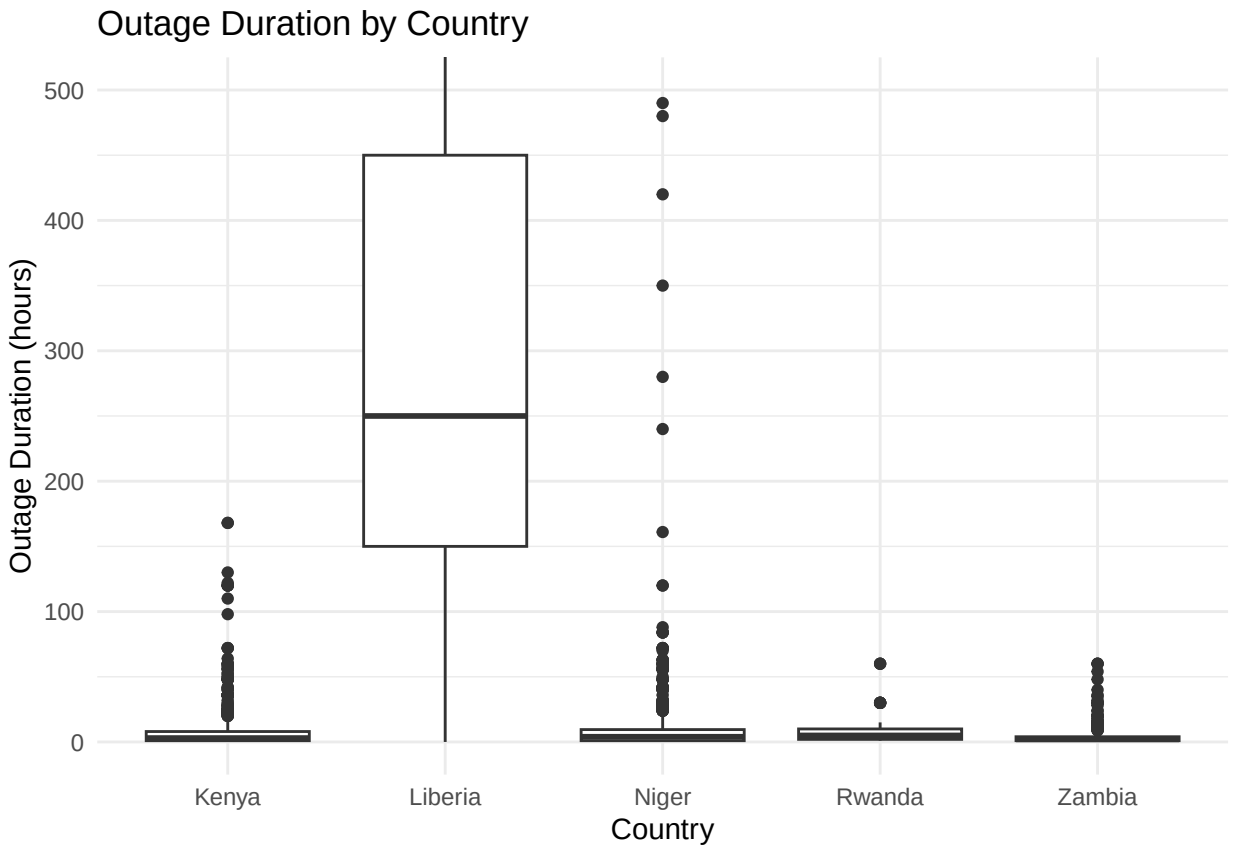
```
ggplot(grid_connection_sumary,
       aes(x=reorder(Country, Connection_Rate),
           y= Connection_Rate)) +
  geom_col(fill="steelblue") +
  labs(
    title = "Household Grid Connection Rates by Country",
    x = "Country",
    y = "Grid-connected households (%)"
  ) +
  theme_minimal()
```



Analysis 2: Outage

```
outage_by_country <- All_Country %>%
  filter(
    !is.na(OutageDurationHours),
    OutageDurationHours >= 0
  )

ggplot(outage_by_country, aes(x = Country, y = OutageDurationHours)) +
  geom_boxplot() +
  labs(
    title = "Outage Duration by Country",
    x = "Country",
    y = "Outage Duration (hours)"
  ) +
  coord_cartesian(ylim = c(
    0,
    quantile(All_Country$OutageDurationHours, 0.95, na.rm = TRUE)
  )) +
  theme_minimal()
```



The boxplot shows differences in the distribution of reported electricity outage durations across countries. Liberia has a notably higher median outage duration and a much wider distribution than the other countries, suggesting that the households in this sample report longer and more variable outages. Kenya and Niger have lower median outage durations but contain high-value outliers. Rwanda and Zambia show comparitively lower and less variable distributions.

Analysis 3: Consumption

```
# Electricity consumption by country
consumption_data <- All_Country %>%
  filter(
    !is.na(ElectricityConsumptionkWh),
    ElectricityConsumptionkWh >= 0
  )

consumption_summary <- consumption_data %>%
  group_by(Country) %>%
  summarise(
    Total_Valid = n(),
    Mean_Consumption = round(mean(ElectricityConsumptionkWh), 2),
    Median_Consumption = round(median(ElectricityConsumptionkWh), 2),
    Max_Consumption = round(max(ElectricityConsumptionkWh), 2)
  )

consumption_summary
```

```
## # A tibble: 5 x 5
```

```
##   Country Total_Valid Mean_Consumption Median_Consumption Max_Consumption
##   <chr>         <int>            <dbl>              <dbl>           <dbl>
## 1 Kenya          1971             63.2               50.6           1848.
## 2 Liberia        3708              1.56               1               111
## 3 Niger           405           2120.                86            67115
## 4 Rwanda         1478             20.2                9.3            400
## 5 Zambia          229            286.                195            14080
```

The table shows considerable variation in total household electricity consumption across five countries. Kenya has a median consumption of 50.6 kWh while Zambia has the highest median comsumption of 195 kWh. Rwanda and Liberia have considerably lower median consumption, at 9.3 kWh and 1 kWh, respectively. Niger has a median of 86 kWh but a much higher mean consumption of 2120 kWh, suggesting that a small number of extremely high consumption observations are skewing iiits average.

Overall Interpretation

Taken together, these figures and preliminary analyses provide an initial picture of differences in household electricity access, consumption, and reliability across the five countries. However, these are preliminary descriptive comparisons and do not establish relationships between electricity access and economic development. Missing observations, extreme values, differences in sample sizes, and potentially inconsistent reporting across countries may influence the results. Further analysis should examine household-level averages and medians and investigate outliers